

Performance Intelligence: Decoding Kimia Farma's Business Performance (2020–2023)

Kimia Farma - Big Data Analytics

Presented by
Zafirah Aida Adista

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Aspiring Data Analyst

An engineering graduate and highly dedicated **Comprehensive Data Analytics** alumnus from **Hacktiv8**. Leveraging a strong engineering background, I possess robust analytical and technical foundations in building **end-to-end data pipelines**, as well as processing complex datasets using **SQL, Python, Spark, and Apache Airflow**. Proficient in conducting **statistical analysis** and visualizing clear business insights through **Power BI, Looker Studio, and Tableau**. I am a fast learner with a **data-driven mindset**, committed to ensuring data accuracy and **transforming raw data into impactful strategic decisions**.



Surabaya



zafirah.adistaa@gmail.com



Zafirah Aida Adista

Courses and Certification

Hacktiv8	-	Comprehensive	Data Analytics		April	2026
Boston Consulting Group	-	Data Science	Job Simulation		April	2026
HackerRank	-	SQL	(Advanced)		April	2026
HackerRank	-	Python	(Basic)		April	2026
Quantium - Data Analytics Job Simulation						April 2026

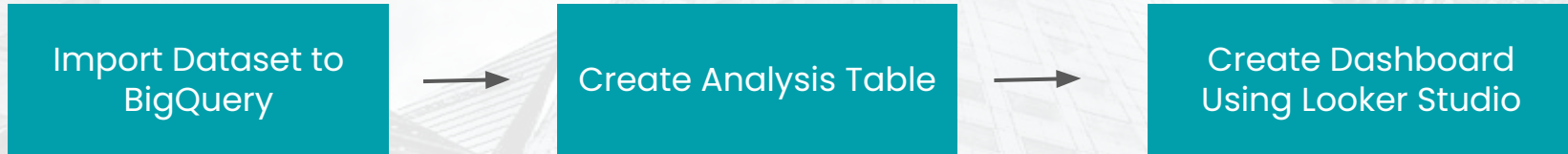
About Company

PT Kimia Farma Tbk is the pioneer and oldest pharmaceutical company in Indonesia, operating as a key member of the State-Owned Enterprise (BUMN) Pharmaceutical Holding. The company delivers fully integrated healthcare services, spanning from research and development and active pharmaceutical ingredient (API) manufacturing to nationwide distribution. Supported by thousands of pharmacies, clinical laboratories, and healthcare clinics, Kimia Farma's network stretches from major urban centers to remote areas. Today, the company is actively transforming into an adaptive digital healthcare solution provider, committed to ensuring accessible, affordable, and high-quality health services for all Indonesians.



Project Portfolio

This project evaluates Kimia Farma's operational and financial performance (2020–2023) by resolving the challenge of **scattered data silos**. By extracting and consolidating fragmented raw transaction data into a **centralized, unified reporting system**, this project eliminated data friction and established a **single source of truth**. The resulting interactive dashboard transforms complex, disconnected datasets into streamlined, actionable insights, empowering management to make swift, **data-driven strategic decisions**.



Available Datasets:

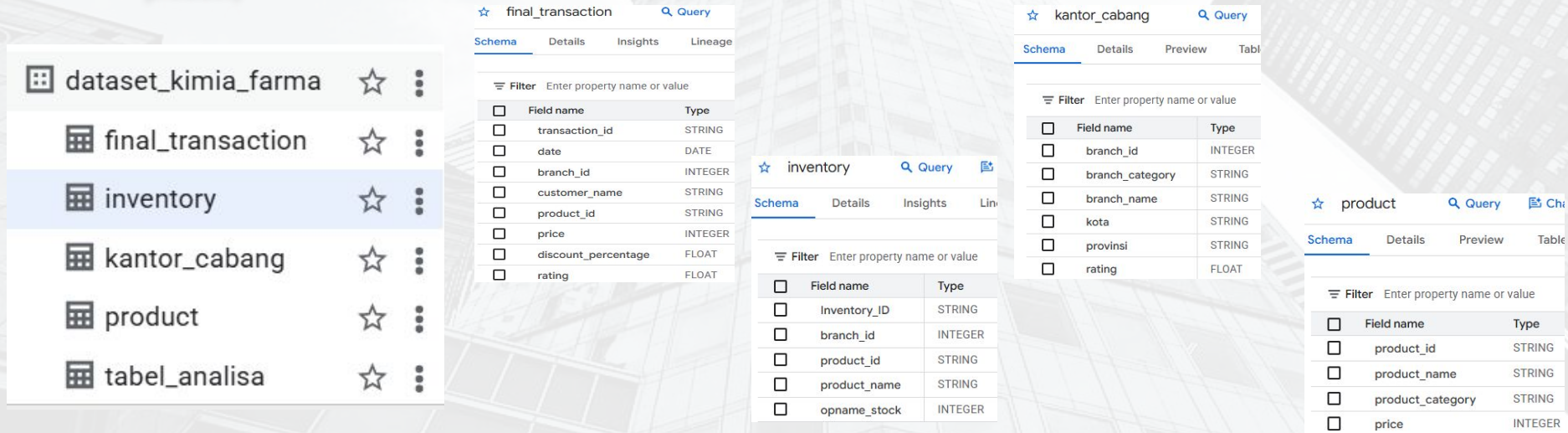
- **kf_transaction:** Daily sales transaction details and activities
- **kf_inventory:** Stock levels, inventory management, and product availability data
- **kf_product:** Complete medicine catalog, including individual unit pricing
- **kf_kantor cabang:** Geographic information and performance ratings for each branch unit across Indonesia

[Project explanation video here!](#)

1. Importing Dataset to BigQuery

The raw CSV datasets were imported into Google Cloud Platform (GCP) BigQuery through the following steps:

1. **Create Dataset:** Initialized a new BigQuery project and created a centralized dataset named kimia_farma.
2. **Configure Table:** Selected the 'Create Table' option within the dataset to prepare for data loading.
3. **Upload & Map:** Uploaded the raw CSV files, set the source format, and specified the destination paths.
4. **Finalize Tables:** Successfully ingested and named each individual table (final_transaction, inventory, kantor_cabang, and product)



The screenshot displays the Google Cloud BigQuery interface. On the left, a sidebar lists the dataset 'dataset_kimia_farma' and its tables: 'final_transaction', 'inventory' (highlighted), 'kantor_cabang', 'product', and 'tabel_analisa'. The main area shows the schema for four tables: 'final_transaction', 'inventory', 'kantor_cabang', and 'product'.

final_transaction Schema:

Field name	Type
transaction_id	STRING
date	DATE
branch_id	INTEGER
customer_name	STRING
product_id	STRING
price	INTEGER
discount_percentage	FLOAT
rating	FLOAT

inventory Schema:

Field name	Type
Inventory_ID	STRING
branch_id	INTEGER
product_id	STRING
product_name	STRING
opname_stock	INTEGER

kantor_cabang Schema:

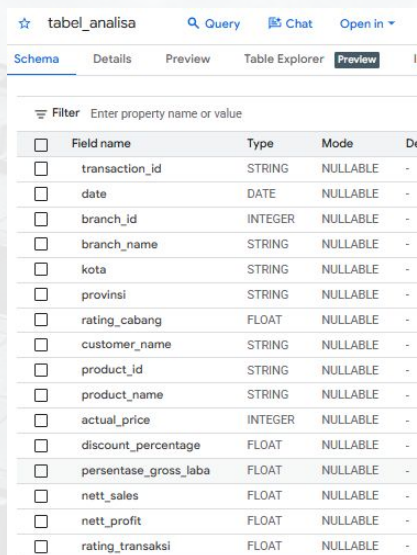
Field name	Type
branch_id	INTEGER
branch_category	STRING
branch_name	STRING
kota	STRING
provinsi	STRING
rating	FLOAT

product Schema:

Field name	Type
product_id	STRING
product_name	STRING
product_category	STRING
price	INTEGER

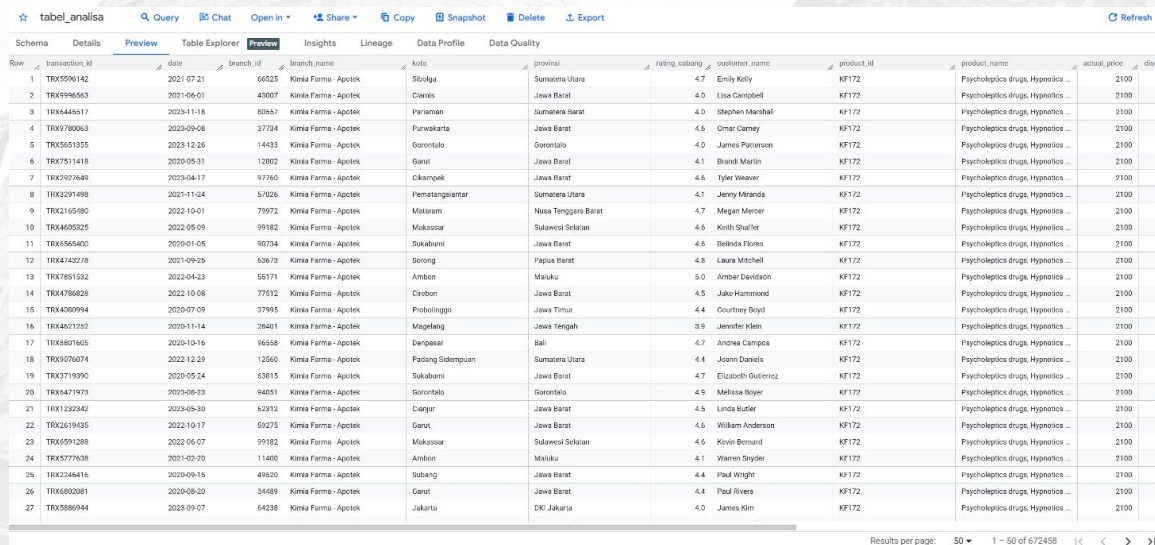
2. Create Analysis Table

Following the successful ingestion of the datasets, the next critical phase involved **Data Transformation**. A centralized master table, designated as **tabel_analisa**, was engineered by consolidating primary data from the transaction, product, and branch tables utilizing **advanced SQL joins**. The primary objective of this integration was to establish a **comprehensive and unified dataset**, thereby **optimizing query performance** and ensuring seamless accuracy within the **Looker Studio visualization layer**.



The screenshot shows the 'tabel_analisa' table schema in Looker Studio. The table has 14 columns: transaction_id, date, branch_id, branch_name, kota, provinsi, rating_cabang, customer_name, product_id, product_name, actual_price, discount_percentage, persentase_gross_laba, nett_sales, nett_profit, and rating_transaksi. The 'Preview' tab is selected, showing a list of rows with their respective values.

Field name	Type	Mode	Default
transaction_id	STRING	NULLABLE	-
date	DATE	NULLABLE	-
branch_id	INTEGER	NULLABLE	-
branch_name	STRING	NULLABLE	-
kota	STRING	NULLABLE	-
provinsi	STRING	NULLABLE	-
rating_cabang	FLOAT	NULLABLE	-
customer_name	STRING	NULLABLE	-
product_id	STRING	NULLABLE	-
product_name	STRING	NULLABLE	-
actual_price	INTEGER	NULLABLE	-
discount_percentage	FLOAT	NULLABLE	-
persentase_gross_laba	FLOAT	NULLABLE	-
nett_sales	FLOAT	NULLABLE	-
nett_profit	FLOAT	NULLABLE	-
rating_transaksi	FLOAT	NULLABLE	-



The screenshot shows the 'tabel_analisa' table data in Looker Studio. The table has 14 columns: transaction_id, date, branch_id, branch_name, kota, provinsi, rating_cabang, customer_name, product_id, product_name, actual_price, discount_percentage, persentase_gross_laba, nett_sales, nett_profit, and rating_transaksi. The 'Preview' tab is selected, showing a list of rows with their respective values.

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id	product_name	actual_price	discount
1	TXK5596142	2021-07-21	66525	Kimia Farma - Apotek	Sibolga	Sumatera Utara	4.7	Emily Kelly	KF172	Psychophleics drugs, Hypnotics ...	2100	
2	TXK9996363	2021-06-01	43007	Kimia Farma - Apotek	Cianjur	Jawa Barat	4.0	Lisa Campbell	KF172	Psychophleics drugs, Hypnotics ...	2100	
3	TXK4444817	2023-11-18	80867	Kimia Farma - Apotek	Pariaman	Sumatera Barat	4.0	Stephen Marshall	KF172	Psychophleics drugs, Hypnotics ...	2100	
4	TXK9780063	2023-09-08	37784	Kimia Farma - Apotek	Purwokarta	Jawa Barat	4.6	Omar Carney	KF172	Psychophleics drugs, Hypnotics ...	2100	
5	TXK5651355	2023-12-26	14433	Kimia Farma - Apotek	Gorontalo	Gorontalo	4.0	James Patterson	KF172	Psychophleics drugs, Hypnotics ...	2100	
6	TXK7511418	2020-05-31	12802	Kimia Farma - Apotek	Garut	Jawa Barat	4.1	Brandi Martin	KF172	Psychophleics drugs, Hypnotics ...	2100	
7	TXK7927648	2023-04-17	97760	Kimia Farma - Apotek	Cikampek	Jawa Barat	4.6	Tyler Weaver	KF172	Psychophleics drugs, Hypnotics ...	2100	
8	TXK3291498	2021-11-24	57026	Kimia Farma - Apotek	Pematangsiantar	Sumatera Utara	4.1	Jenny Miranda	KF172	Psychophleics drugs, Hypnotics ...	2100	
9	TXK2165480	2022-10-01	79972	Kimia Farma - Apotek	Mataram	Nusa Tenggara Barat	4.7	Megan Mercer	KF172	Psychophleics drugs, Hypnotics ...	2100	
10	TXK4605325	2022-05-09	99182	Kimia Farma - Apotek	Makassar	Psychophleics drugs, Hypnotics ...	4.6	Koith Shaf'rol	KF172	Psychophleics drugs, Hypnotics ...	2100	
11	TXK5660400	2020-01-05	50704	Kimia Farma - Apotek	Bukabuni	Jawa Barat	4.6	Balinda Flores	KF172	Psychophleics drugs, Hypnotics ...	2100	
12	TXK4743278	2021-09-26	63673	Kimia Farma - Apotek	Sorong	Papua Barat	4.8	Laura Mitchell	KF172	Psychophleics drugs, Hypnotics ...	2100	
13	TXK7851532	2022-04-23	56171	Kimia Farma - Apotek	Ambon	Maluku	5.0	Amber Davidson	KF172	Psychophleics drugs, Hypnotics ...	2100	
14	TXK4786828	2022-10-08	77512	Kimia Farma - Apotek	Cirebon	Jawa Barat	4.5	Jake Hammond	KF172	Psychophleics drugs, Hypnotics ...	2100	
15	TXK4009994	2020-07-09	37995	Kimia Farma - Apotek	Probolinggo	Jawa Timur	4.4	Courtney Boyd	KF172	Psychophleics drugs, Hypnotics ...	2100	
16	TXK4621292	2020-11-14	28401	Kimia Farma - Apotek	Magelang	Jawa Tengah	3.9	Jennifer Klein	KF172	Psychophleics drugs, Hypnotics ...	2100	
17	TXK8871605	2020-10-16	96568	Kimia Farma - Apotek	Dempasar	Bali	4.7	Andrea Campos	KF172	Psychophleics drugs, Hypnotics ...	2100	
18	TXK9076074	2022-12-29	12560	Kimia Farma - Apotek	Padang Sidempuan	Sumatera Utara	4.4	Joann Daniels	KF172	Psychophleics drugs, Hypnotics ...	2100	
19	TXK3719990	2020-05-24	63815	Kimia Farma - Apotek	Sukabuni	Jawa Barat	4.7	Elizabeth Gutierrez	KF172	Psychophleics drugs, Hypnotics ...	2100	
20	TXK4471973	2023-09-23	94031	Kimia Farma - Apotek	Gorontalo	Gorontalo	4.9	Melissa Boyer	KF172	Psychophleics drugs, Hypnotics ...	2100	
21	TXK1232282	2023-05-30	62912	Kimia Farma - Apotek	Cianjur	Jawa Barat	4.5	Linda Butler	KF172	Psychophleics drugs, Hypnotics ...	2100	
22	TXK2619435	2022-10-17	59275	Kimia Farma - Apotek	Garut	Jawa Barat	4.6	William Anderson	KF172	Psychophleics drugs, Hypnotics ...	2100	
23	TXK6591288	2022-04-07	99182	Kimia Farma - Apotek	Makassar	Sulawesi Selatan	4.6	Kevin Bonnard	KF172	Psychophleics drugs, Hypnotics ...	2100	
24	TXK5776838	2021-05-29	11400	Kimia Farma - Apotek	Ambon	Maluku	4.1	Warren Snyder	KF172	Psychophleics drugs, Hypnotics ...	2100	
25	TXK12346416	2020-06-16	49620	Kimia Farma - Apotek	Subang	Jawa Barat	4.4	Paul Wright	KF172	Psychophleics drugs, Hypnotics ...	2100	
26	TXK6802081	2020-08-20	34089	Kimia Farma - Apotek	Garut	Jawa Barat	4.4	Paul Rivers	KF172	Psychophleics drugs, Hypnotics ...	2100	
27	TXK5886944	2023-09-07	64228	Kimia Farma - Apotek	Jakarta	DKI Jakarta	4.0	James Kim	KF172	Psychophleics drugs, Hypnotics ...	2100	

2. Create Analysis Table

Big Query Syntax

```
1 CREATE OR REPLACE TABLE `kimia-farma-internship-497611.dataset_kimia_farma.tabel_analisa` AS
2 SELECT
3     t.transaction_id,
4     t.date,
5     t.branch_id,
6     c.branch_name,
7     c.kota,
8     c.provinsi,
9     c.rating AS rating_cabang,
10    t.customer_name,
11    t.product_id,
12    p.product_name,
13    t.price AS actual_price,
14    t.discount_percentage,
```

- Implemented **DDL (Data Definition Language) syntax** using **CREATE OR REPLACE TABLE** to ensures the analysis table is **automatically updated** whenever raw data updates occur
- **Consolidated primary attributes** from transaction, product, and kantor_cabang

```
16 -- 1. Menghitung persentase gross laba berdasarkan ketentuan harga
17 CASE
18     WHEN t.price <= 50000 THEN 0.10
19     WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
20     WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
21     WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
22     ELSE 0.30
23 END AS persentase_gross_laba,
```

- Deployed **CASE WHEN** logic to calculate profit margins dynamically based on product price
- Replaced manual entry with an **automated system**, ensuring **100% accuracy** across margin tiers (10%–30%).

2. Create Analysis Table

Big Query Syntax

```
25 -- 2. Menghitung nett_sales (harga setelah dipotong diskon)
26 (t.price * (1 - (t.discount_percentage / 100))) AS nett_sales,
```

Automatically calculated **nett_sales** (price after discount) and **nett_profit** (net profit) for each individual transaction.

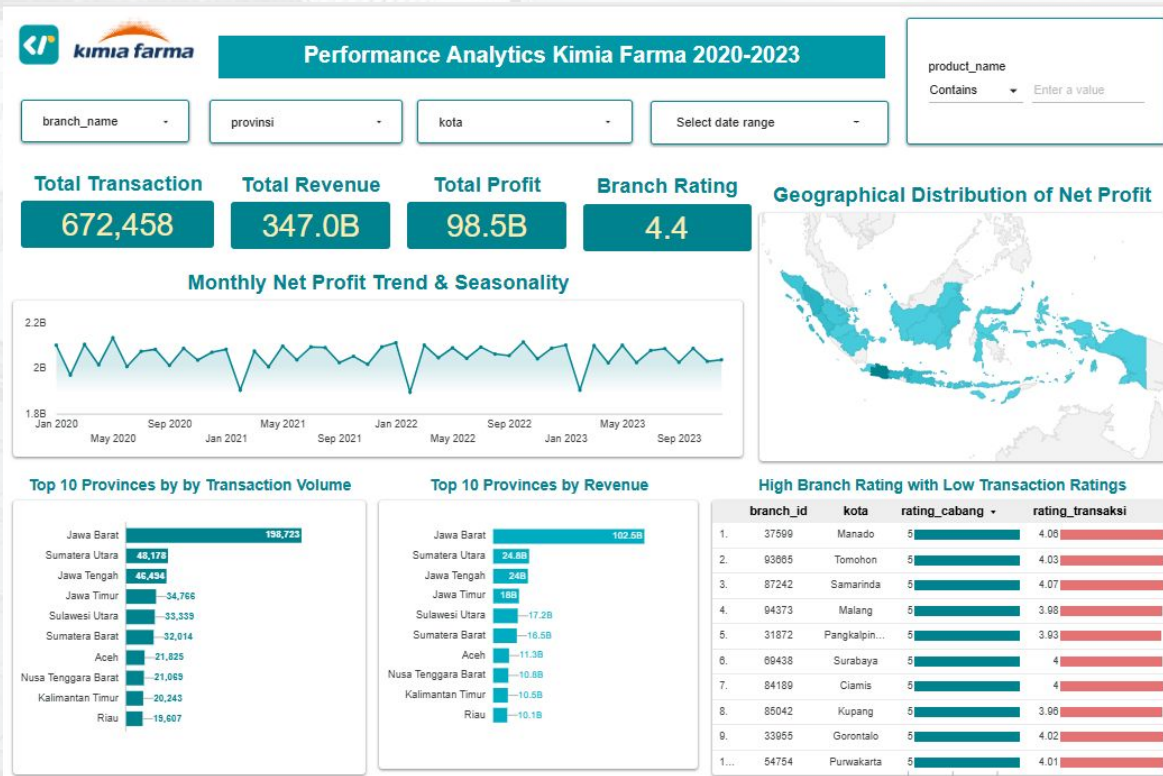
```
28 -- 3. Menghitung nett_profit (nett_sales dikali persentase laba)
29 ((t.price * (1 - (t.discount_percentage / 100))) * CASE
30   WHEN t.price <= 50000 THEN 0.10
31   WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
32   WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
33   WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
34   ELSE 0.30
35 END
36 ) AS nett_profit,
37
38 t.rating AS rating_transaksi
```

Formulated a nested calculation by multiplying the calculated **nett_sales** with the dynamic **CASE WHEN profit tiers**

```
40 FROM
41   `kimia-farma-internship-497611.dataset_kimia_farma.final_transaction` t
42 LEFT JOIN
43   `kimia-farma-internship-497611.dataset_kimia_farma.kantor_cabang` c
44   ON t.branch_id = c.branch_id
45 LEFT JOIN
46   `kimia-farma-internship-497611.dataset_kimia_farma.product` p
47   ON t.product_id = p.product_id;
```

Executed **LEFT JOIN** operations to map the transaction log to geographic branches and product catalogs using **unique foreign keys** (**branch_id** and **product_id**)

3. Dashboard Performance Analytics



OUTCOME :

- **Profit Trends:**
Visualizing monthly net profit and seasonality
- **Geographic Reach:**
Identifying top-performing regions by transaction volume
- **Anomaly Detection:**
Gap analysis between branch ratings and transaction quality.

[VIEW FULL DASHBOARD HERE](#)

3. Insight & Recommendation

Insight:

- **Revenue Engine:**
Jawa Barat dominates the market, driving the highest transaction volume (~150K) and revenue (~Rp 107B)
- **Predictable Seasonality:**
Net profit tracks a clear cyclical trend, consistently **peaking in March and September**, but experiencing **sharp drops every January** (post-holiday slump) and a **mid-year compression (May–July)**
- **Operational Paradox:**
Key branches (e.g., **Manado, Tomohon, Samarinda**) maintain a perfect internal **Branch Rating of 5.0**, yet suffer from low customer **Transaction Ratings (~4.0)**

Recommendation:

- **Scale Success Blueprint:**
Replicate **Jawa Barat's** operational and marketing strategies in other high-potential regions like **Sumatera Utara** and **Jawa Tengah**
- **Counter-Seasonality Campaigns:**
Launch targeted promotions in **late December/January** and **April/May** to stabilize cash flow and mitigate predictable revenue valleys
- **Targeted Branch Audit:**
Conduct an immediate service and supply chain audit on the **perfectly-rated internal branches** that underperform in customer satisfaction

Thank You



Rakamin
Academy



Kimia Farma

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